DATA SCIENCE PROJECT REPORT

App User Behavior Segmentation

Using Unsupervised Machine Learning

Project domain

User Behavior Analytics • Machine Learning

A practical study of app engagement patterns, user grouping, and data-driven retention opportunities.

Prepared from the supplied project brief, dataset, and preprocessing notebook

Executive Summary

This project studies app user activity to understand how people engage with an application. The central challenge is that user segments are not supplied in advance. Instead of predicting a known label, the project brief proposes an unsupervised learning approach to discover groups of users with similar behavioral patterns.

The supplied raw dataset contains **50,000 rows and 25 columns**. It includes demographic information, device and subscription details, usage activity, engagement indicators, and a churn-risk score. The preprocessing notebook documents missing-value treatment, duplicate checking, IQR-based outlier detection, percentile capping, skewness inspection, and Yeo-Johnson transformation.

The project brief reports a K-Means solution with four behavioral segments: high engagement, moderate engagement, low engagement / at-risk, and occasional users. It also describes PCA visualization as a way to inspect separation between the groups. The business purpose is to support more relevant engagement, retention, loyalty, and product decisions.

Evidence note: The uploaded notebook contains preprocessing steps but does not include the K-Means fitting code, elbow plot, PCA output, cluster counts, or cluster evaluation values. Therefore, the four segment descriptions below are presented as reported in the project brief, not as independently recalculated results from the supplied notebook.

Project at a Glance

Item	Details
Dataset records	50,000
Dataset columns	25
Numerical columns	20
Categorical columns	5
Missing values in rating_given	5,022 (10.04%)
Duplicate rows detected	0
Average age	38.5 years
Average sessions per week	8.0
Average engagement score	64.94
Average churn risk score	0.501
Most common device	Android (27,315 users)

Project Objectives

• Discover groups of users with similar app usage and engagement behavior.

• Understand which users are highly engaged, moderately engaged, inactive, or irregular in their usage.

• Translate behavioral patterns into practical business actions.

• Support retention planning, personalized experiences, and product improvement.

1. Business Problem and Project Scope

Applications collect large volumes of activity data, but raw event and usage records do not automatically explain what different users need. A single generic campaign may not work equally well for highly active users, occasional users, and people showing signs of disengagement.

This project aims to group users according to similarities in their behavior. Since predefined segment labels are not available, the project brief uses unsupervised machine learning to discover the groups from the data.

Business Use Cases

Use case	How segmentation can help
Targeted marketing	Create campaigns that reflect each group's engagement and usage patterns.
Churn-risk identification	Prioritize low-activity users for timely retention communication.
Personalized experience	Tailor recommendations, notifications, and feature guidance to user needs.
Feature optimization	Compare how segments interact with app features and identify improvement opportunities.
Data-driven decisions	Give product, marketing, and support teams a clearer view of user behavior.

Scope and interpretation

The supplied brief focuses on user-level segmentation. It identifies sessions, session duration, engagement score, and activity metrics as relevant behavioral inputs. Demographic and categorical fields may provide useful context for profiling, but categorical values must be encoded appropriately before they can be used in a distance-based clustering model.

A churn-risk score exists in the raw dataset. If it is used only after clustering to describe or compare segments, it can support interpretation. If it is included as a clustering input, the resulting groups may partly reflect that pre-existing score. This choice should be stated clearly in the final implementation.

2. Dataset Understanding

The raw CSV has **50,000 user records** and **25 fields**. Each row represents one app user. The columns cover identity, demographics, device, usage, engagement, subscription, feedback, and acquisition source.

Feature groups

Feature group	Examples	Potential role
User and demographic	user_id, age, gender, country	Identification and user profile
Device and account	device_type, app_version, account_age_days	Context for usage differences
Session activity	sessions_per_week, avg_session_duration_min, daily_active_minutes	Core engagement behavior
Feature interaction	feature_clicks_per_session, in_app_search_count, pages_viewed_per_session	Depth of app interaction
Communication and feedback	notifications_opened_per_week, rating_given, support_tickets_raised	Response and experience indicators
Retention indicators	days_since_last_login, churn_risk_score	Recency and risk context
Monetization / sharing	subscription_type, ads_clicked_last_30_days, content_downloaded, social_shares	Value and interaction context
Acquisition	marketing_source	How users entered the app

Selected dataset profile

Metric	Observed value
Dataset records	50,000
Dataset columns	25
Numerical columns	20
Categorical columns	5
Missing values in rating_given	5,022 (10.04%)
Duplicate rows detected	0
Average age	38.5 years

Top country values

Country	Users
India	22,550
USA	9,115
UK	5,043
Canada	3,853
Australia	3,498

Subscription mix

Subscription type	Users
Free	30,088
Basic	12,495
Premium	7,417

Data quality observation

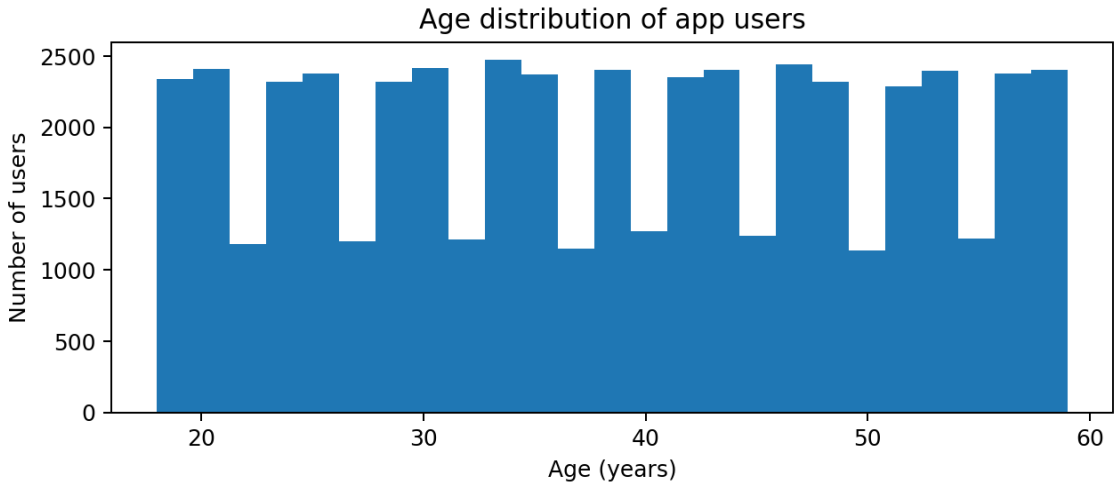
The raw dataset contains 5,022 missing values in rating_given (10.04% of rows). The other columns have no missing values in the supplied raw CSV. The notebook fills rating_given with its median. The notebook also checks for duplicate rows; the raw dataset has 0 exact duplicate rows.

The dataset includes a user_id field. It should be retained for traceability and customer-level reporting, but excluded from distance calculations because an identifier is not a behavioral measurement.

3. Exploratory Data Analysis

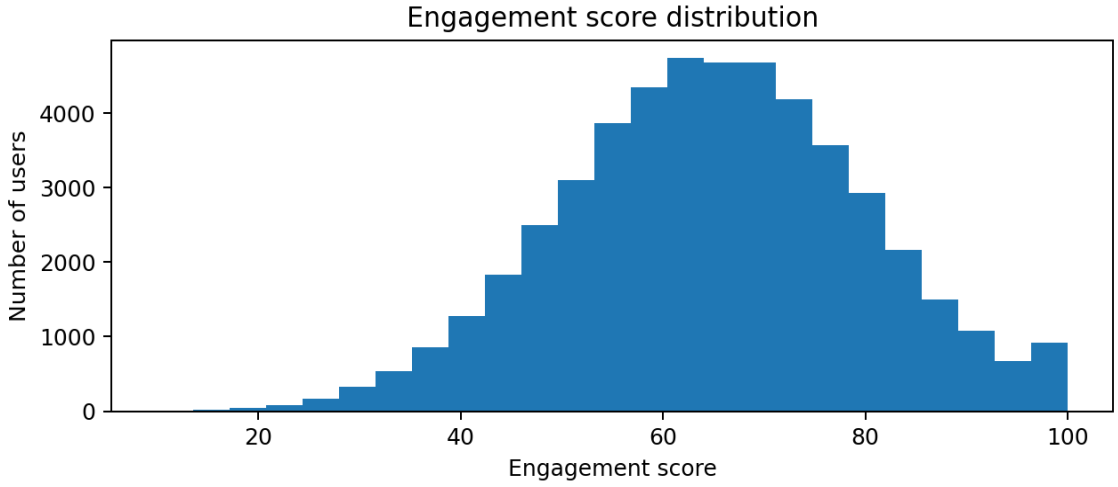
EDA helps understand distributions, ranges, and data quality before building a clustering model. It is also useful for spotting variables that may dominate distance calculations or need transformation.

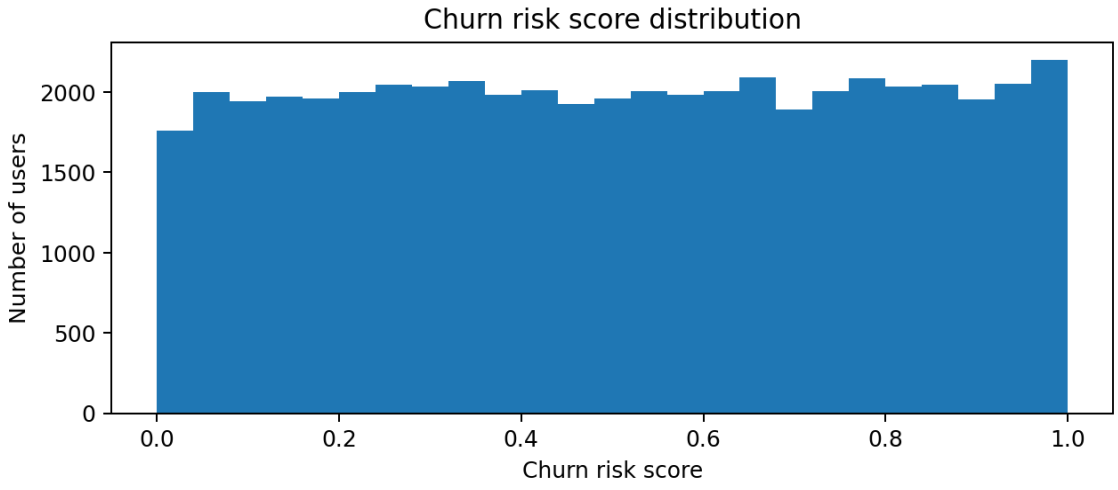
Age distribution



The average age in the supplied dataset is 38.5 years. The age field ranges from 18 to 59 years. Age can help describe a segment, but it should not automatically be treated as the main driver of behavioral grouping.

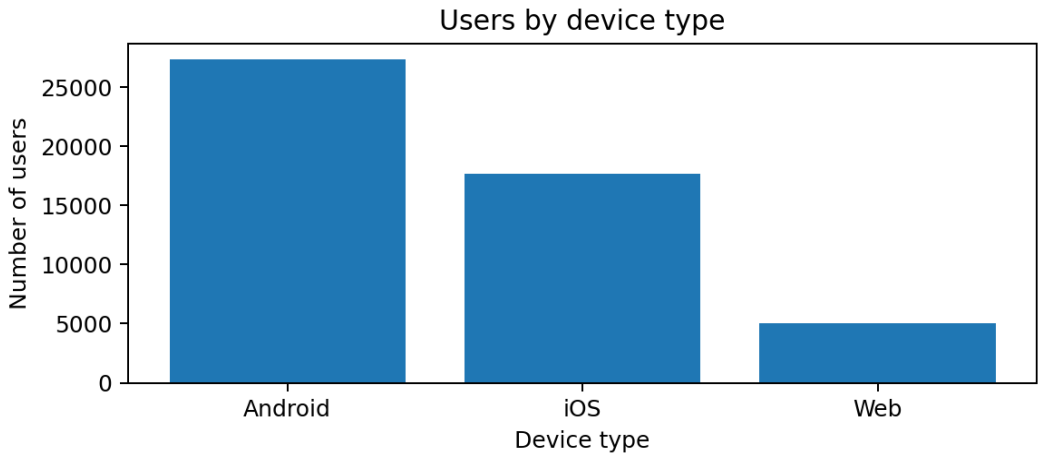
Engagement and churn indicators





The average engagement score is 64.94, while the average churn-risk score is 0.501. These are dataset-level averages; they do not represent the average of any individual cluster.

Device distribution



EDA interpretation

• Review numeric distributions and ranges before scaling.

• Use categorical frequency tables to understand the population mix.

• Check missingness, duplicates, unusual values, and skewness.

• Compare cluster profiles only after cluster labels have been generated.

4. Data Cleaning and Preprocessing

The supplied preprocessing notebook follows a sequence of data-quality checks and transformations. The steps below reflect the code present in the uploaded notebook.

4.1 Missing-value imputation

The notebook identifies missing values using isnull().sum() and fills missing rating_given values with the column median. Median imputation is less affected by extreme values than mean imputation. In the raw CSV, rating_given has 5,022 missing entries, approximately 10.04%.

4.2 Duplicate and categorical checks

The notebook checks exact duplicate rows using duplicated(). It also prints value counts for object-type columns to inspect category consistency. The raw file has no exact duplicate rows.

4.3 Outlier detection and capping

The notebook first uses the IQR rule: values below Q1 – 1.5 × IQR or above Q3 + 1.5 × IQR are flagged for inspection. It then visualizes selected numeric columns with boxplots. Rather than deleting observations, it caps selected columns at their 1st and 99th percentiles using clip().

The notebook applies percentile capping to sessions_per_week, avg_session_duration_min, daily_active_minutes, feature_clicks_per_session, notifications_opened_per_week, in_app_search_count, crash_events_last_30_days, support_tickets_raised, content_downloads, social_shares, ads_clicked_last_30_days, and engagement_score.

4.4 Skewness and transformation

The notebook checks skewness and applies PowerTransformer(method='yeo-johnson') to crash_events_last_30_days, avg_session_duration_min, social_shares, feature_clicks_per_session, rating_given, and support_tickets_raised. Yeo-Johnson can handle zero and negative values, unlike a basic logarithmic transformation.

4.5 Final checks and export

The notebook checks the final shape, info, descriptive statistics, and sample rows, then saves the processed data as app_user_behavior_preprocess.csv. That exported file was not present among the mounted files available for this report, so its final values could not be independently inspected.

Important implementation check

The notebook's numeric-column list is created before PowerTransformer is applied. That is acceptable for the current set of numeric columns, but the final model pipeline should explicitly select features, exclude user_id, and ensure all input columns are numeric and scaled. Fit imputation, transformation, and scaling on training data only when using a train/test evaluation design.

5. Unsupervised Learning and K-Means

Unsupervised learning is used when the dataset does not contain a known target label for each observation. The algorithm looks for structure or similarity in the input data. In this project, the intended output is a set of user groups based on behavioral patterns.

Why K-Means?

• It is a widely used clustering algorithm for grouping numeric observations.

• It assigns each user to the nearest cluster center.

• It is relatively straightforward to explain and can scale to a dataset of this size when features are prepared properly.

• The resulting groups can be profiled using average engagement and activity metrics.

How K-Means works

Step	Explanation
1. Choose K	Set the number of clusters to form.
2. Initialize centroids	Start with K cluster-center positions.
3. Assign users	Assign each user to the nearest centroid using distance.
4. Update centroids	Recalculate each centroid as the mean of its assigned users.
5. Repeat	Continue assignment and update until the solution stabilizes.

Why scaling matters

K-Means commonly uses Euclidean distance. If one feature has a much larger numeric range than another, it can dominate the distance calculation. StandardScaler transforms a feature using its mean and standard deviation so that features are on comparable scales. The project brief states that StandardScaler was used before clustering.

Choosing the number of clusters: Elbow Method

The Elbow Method fits K-Means for different values of K and records inertia, also called within-cluster sum of squares. Inertia usually decreases as K increases. The elbow is the point after which adding more clusters produces a smaller improvement. The project brief says this method was used to select four clusters; the actual elbow chart and inertia values were not included in the supplied notebook.

PCA for visualization

Principal Component Analysis (PCA) reduces a high-dimensional feature space into a smaller number of components. A two-component projection can make it easier to visualize user groups on a scatter plot. PCA is a visualization aid; it does not by itself prove that clusters are valid or commercially meaningful.

6. Reported Segments and Business Interpretation

The project brief reports that 50,000 users were assigned to four behavioral clusters. The segment names and descriptions below are taken from that brief.

Reported segment	Behavior described in the project brief	Possible business action
Cluster 0 – High Engagement	High session frequency, longer session duration, high engagement scores, and low churn-risk indicators.	Reward loyalty, offer relevant premium features, and invite feedback.
Cluster 1 – Moderate Engagement	Consistent but average usage patterns with moderate engagement and manageable churn risk.	Encourage deeper feature discovery and maintain regular engagement.
Cluster 2 – Low Engagement / At-Risk	Low activity, infrequent logins, shorter sessions, and higher churn-risk indicators.	Prioritize retention messages, onboarding support, and investigate
Cluster 3 – Occasional Users	Irregular interaction, sporadic usage, and lower feature interaction.	Use timely reminders and simple reactivation journeys.

How to explain the result

The cluster number is only an identifier. Cluster 0 is not automatically better than Cluster 1 because of its number; the interpretation comes from comparing the average feature values of each cluster. The business labels should be assigned after reviewing those profiles.

What should be added to make the results stronger

• A table showing the number and percentage of users in each cluster.

• A cluster profile table with mean or median values for sessions, duration, daily active minutes, engagement, last login, and churn-risk score.

• An elbow plot with the tested K values and inertia values.

• A PCA scatter plot colored by cluster label.

• A quantitative cluster quality measure such as silhouette score, reported with the metric's limitations.

The brief describes clear behavioral separation in PCA and says every user received a cluster label. The numerical outputs needed to independently verify these claims were not present in the uploaded preprocessing notebook.

7. Evaluation, Limitations and Future Work

Evaluation without ground-truth labels

Because this is an unsupervised learning task, ordinary classification accuracy is not an appropriate primary metric. There is no known correct segment label to compare against. The project brief provides a project evaluation rubric covering data quality, feature selection and scaling, cluster quality, interpretability, business relevance, and scalability.

Evaluation area	Weight in supplied brief	Evidence to show
Data quality and preprocessing accuracy	20%	Missing-value summary, duplicate check, outlier handling and final data checks.
Feature selection and scaling	15%	Selected feature list and StandardScaler evidence.
Cluster quality and separation	25%	Elbow plot, PCA visualization and optional silhouette score.
Cluster interpretability	15%	Cluster profile table and understandable segment labels.
Business relevance and actionability	15%	Action plan linked to each user segment.
Scalability and performance	10%	Runtime, dataset size, and practical deployment considerations.

Limitations to acknowledge

• The uploaded notebook covers preprocessing but does not show the complete clustering and evaluation code.

• The final processed CSV was not available for independent validation in this report.

• The four segment descriptions are qualitative; exact cluster sizes and centroid statistics were not supplied.

• K-Means is sensitive to feature selection, scaling, initialization, and the chosen K value.

• The data appears to be a project dataset; business conclusions should be validated against live app behavior before operational use.

Future improvements

• Build a reproducible end-to-end pipeline that includes preprocessing, scaling, K-Means, and saved cluster assignments.

• Compare K-Means with alternatives such as hierarchical clustering or DBSCAN where appropriate.

• Evaluate stability across random seeds and time periods.

• Build a dashboard that allows teams to inspect segment sizes and behavior.

• Track whether retention campaigns improve activity and reduce churn over time.

Conclusion

This project demonstrates how app activity data can be prepared and analyzed to understand user behavior. The preprocessing notebook provides a documented foundation, while the project brief describes a four-segment K-Means outcome. The practical value is the move from treating all users the same to designing engagement and retention actions based on observed patterns. The next step is to attach the reproducible clustering code and numerical validation outputs so that the reported segments can be fully audited.

8. Reproducibility Notes and Project Checklist

Files reviewed

• App User Behavior Segmentation.pdf – project brief, objectives, approach, reported results, and evaluation rubric.

• app_user_behavior_dataset.csv – raw dataset used for data profiling in this report.

• App_Preprocessing.ipynb – preprocessing code for missing values, duplicates, outliers, skewness, transformation, and export.

Core preprocessing code documented in the notebook

```
df["rating_given"] = df["rating_given"].fillna(df["rating_given"].median())
```

```
low, high = df[col].quantile([0.01, 0.99])
df[col] = df[col].clip(low, high)
```

```
pt = PowerTransformer(method="yeo-johnson")
df[cols] = pt.fit_transform(df[cols])
```

Before final evaluation or viva

• Confirm the exact numerical feature list used by K-Means.

• Confirm whether churn_risk_score was excluded from clustering and used only for interpretation.

• Show the StandardScaler and K-Means code.

• Show the elbow plot and explain why K = 4 was selected.

• Show the PCA plot and cluster profile/count table.

• Be ready to explain that cluster labels are discovered patterns, not predefined categories.

Prepared as a clear, presentation-oriented report based on the supplied project materials. Any result not evidenced by the uploaded notebook is explicitly identified rather than estimated.